

Shubham Singh

Seattle, WA · 812-778-4049

shubh.singh.dev@gmail.com · linkedin.com/in/shusingh

github.com/shusingh · shusingh.github.io

Software engineer (SDE II) at Amazon with four years building production backend and distributed systems on AWS, now focused on **agentic AI**: RAG on Bedrock, evaluation harnesses, and guardrails that make LLM systems safe to ship.

EXPERIENCE

Amazon

Seattle, WA · Aug 2022 – Present

SDE II (Oct 2025 – Present) · SDE I (Aug 2022 – Oct 2025)

ComplianceIQ: Agentic AI Platform · Bedrock, AgentCore, Strands, MCP

- Led design and delivery of a citation-grounded Q&A platform on Bedrock for 5 partner teams; owned backend and UI architecture and directed **3 junior engineers**, cutting escalated engineering tickets from ~50 to **~5 per month**.
- Extended the agent from answering to acting: built tools over the knowledge base and data catalog systems and integrated downstream MCP servers, so risk managers configure a control launch by stating intent ("clone control X, but change Y") in **minutes instead of hours**, behind audited tool calls and human approval.
- Designed a fail-closed grounding policy (Bedrock Guardrails, Automated Reasoning checks) that declines rather than returns an unsupported claim, cutting unsupported answers from ~45% on the base model to **none detected**.
- Built the evaluation harness that gates every release: a 500+ case suite covering grounding, citation correctness, and refusal behavior, with **99% of answers** passing citation verification in the suite and production.
- Stood up agent-to-agent (A2A) infrastructure so other teams' agents can invoke the Strands agent as a reusable capability.

Regulatory Control Launch Platform · Step Functions, DynamoDB, Java, React, TypeScript

- Owned UI design (HLD and LLD) and delivery for a platform running **2,000+ regulatory control launches per year**, replacing a manual, multi-team process and cutting launch time from 20+ weeks to **under 2 days**.
- Designed and built a Pre-Launch Testing platform that validates each control against **~10 downstream systems** and blocks faulty launches, replacing post-launch investigations that meant chasing 10+ teams.
- Architected a Step Functions workflow orchestrator to replace brittle Lambda chains, running marketplaces in parallel so one failure never blocks the others and each is recorded for risk managers; cut failure rate from **8% to under 1%**.

Compliance Data Pipeline · PySpark, AWS

- Led a regulatory schema migration across **27 data streams** in one week, coordinating ~30 teams with no compliance gap.
- Built a PySpark compliance pipeline that quarantines invalid records instead of failing the batch, with schema versioning and field-level validation across **30M+ records per month**.

PROJECTS & OPEN SOURCE

- Agent Replay** · React, TypeScript, Canvas, Web Workers · 2026
Time-travel debugger for OpenTelemetry-style LLM agent traces: scrub runs like video, replay token streams, and diff two runs. Renders **50k+ spans at 60fps** via structure-of-arrays geometry, segment-tree culling, and LOD aggregation.
- Michi** · Go, React, TypeScript, Groq, MapLibre GL · 2026
Full-stack AI trip planner: a Go serverless backend calls Groq (Llama 3.3 70B) with per-IP token-bucket rate limiting; a React frontend maps results live.
- Giskard** · Python, LLM Evaluation · 2026
Merged JsonValid, a JSON Schema validation check for LLM outputs, into Giskard's open-source eval library (**v3.0.0**).

SKILLS

- Languages:** Python, TypeScript, Java, Go, SQL
- AI & Data:** Bedrock, RAG, BM25/kNN retrieval, OpenSearch, Guardrails, PySpark, Postgres, Redis
- Cloud & Application:** Step Functions, Lambda, DynamoDB, S3, SQS/SNS, EventBridge, CDK, Docker, React, FastAPI

EDUCATION

Indiana University Bloomington: M.S. in Data Science

Jan 2021 – Jul 2022